From Basel to Gauss: Sums, Products and Integrals

Maxprograms
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A short walk through classical results of analysis: telescoping sums, the geometric series, Euler's product for the sine, the Gaussian integral, an eigenvalue problem and the binomial theorem. Each result is stated, proved in a few lines and checked against numbers.

1. Introduction

In 1734 Leonhard Euler found the exact value of the sum of the reciprocals of the squares of the natural numbers, a problem that had resisted the best mathematicians for almost a century:

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \tag{1}$$

The series converges, because its terms behave like $1/n^2$ for large n, but it converges slowly: after n terms the error is about $1/n$, as the table in Section 9 shows. This article collects proofs of nearby facts that use the same tools, from the first year of calculus to a little linear algebra. Equation (1) is the destination of the walk.

2. Telescoping sums

> **Theorem 1.** *For every natural number $n \geq 1$,*
>
> $$\sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{n}{n+1} \tag{2}$$
>
> *Proof.* Each term can be split into two fractions:
>
> $$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1} \tag{3}$$
>
> so the sum telescopes: every term cancels with the next one, and only the first and the last survive. What remains is $1 - \frac{1}{n+1} = n/\big(n+1\big)$. ∎

Letting $n \to \infty$ gives a sum of exactly 1. The same idea bounds the series of (1): for $k \geq 2$ we have $1/k^2 < 1/\big(k\big(k-1\big)\big)$, and the second sum telescopes as above, so

$$\sum_{k=1}^{n} \frac{1}{k^2} < 2 - \frac{1}{n} < 2 \tag{4}$$

The partial sums grow and stay below 2, so the series converges.

3. The geometric series

For a real number $r \neq 1$ the partial sums of the powers of r have a closed form:

$$\sum_{k=0}^{n} r^k = \left(1 - r^{n+1}\right)/(1 - r) \tag{5}$$

To see this, multiply the sum by $1 - r$: all the intermediate powers cancel in pairs and only $1 - r^{n+1}$ is left. When $|r| < 1$ the power r^{n+1} tends to 0, and the series has a sum:

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}, \qquad |r| < 1 \tag{6}$$

With $r = 1/2$ this says that $1 + 1/2 + 1/4 + 1/8 + \cdots$ adds up to 2, the picture of a segment that is halved again and again.

4. Euler's product for the sine

Euler's idea was to treat the sine as a polynomial of infinite degree whose roots are the integers. Written as a product over its roots, and scaled so that its value at 0 is 1, it becomes[1]

$$\frac{\sin(\pi x)}{\pi x} = \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right) \tag{7}$$

On the other hand, the Taylor series of the sine, divided by πx, gives

$$\frac{\sin(\pi x)}{\pi x} = \sum_{k=0}^{\infty} \frac{(-1)^k (\pi x)^{2k}}{(2k+1)!} = 1 - \frac{\pi^2 x^2}{6} + \frac{\pi^4 x^4}{120} - \cdots \tag{8}$$

Compare the coefficients of x^2 in (7) and (8). Multiplying out the product, the only way to get x^2 is to take one factor $-x^2/n^2$ and 1 from all the others, so the coefficient is $-\sum_{n=1}^{\infty} 1/n^2$. In the series it is $-\pi^2/6$. Equating them gives (1).

5. The Gaussian integral

> **Theorem 2.** *The area under the bell curve is*
>
> $$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi} \tag{9}$$

Proof. Call the integral I. The trick is to square it and to read the product of two integrals as an integral over the plane:

$$I^2 = \iint_{\mathbb{R}^2} e^{-\left(x^2 + y^2\right)}\, dx\, dy \tag{10}$$

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1. The product converges for every real x. Taking $x = 1/2$ gives the product of Wallis, $\pi/2 = \prod_{n=1}^{\infty} \frac{4n^2}{4n^2 - 1}$. A rigorous proof of the product needs the factorization theorem of Weierstrass.

In polar coordinates the area element is $r\,dr\,d\theta$, and the integral becomes elementary:

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2}\, r\, dr\, d\theta = 2\pi \cdot \tfrac{1}{2} = \pi \tag{11}$$

Since I is positive, $I = \sqrt{\pi}$. ∎

Statisticians use this result every day. Dividing the bell curve by its area, and allowing a mean μ and a standard deviation σ, gives the density of the normal distribution, whose total probability is 1 as it should be:

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx = 1 \tag{12}$$

6. A small eigenvalue problem

Linear algebra has its own sums. Take the symmetric matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \tag{13}$$

Its eigenvalues λ are the roots of the characteristic polynomial:

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 \tag{14}$$

The quadratic formula gives the roots of any polynomial $ax^2 + bx + c$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{15}$$

For (14) it gives $\lambda = (4 \pm 2)/2$, that is, $\lambda = 1$ and $\lambda = 3$. A check with the vector $(1, 1)^T$ confirms the larger one:

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{16}$$

In the same way $(1, -1)^T$ is an eigenvector for $\lambda = 1$. The two eigenvectors are orthogonal, as they must be for a symmetric matrix.

7. The binomial theorem

Expanding the power of a sum gives the coefficients that Pascal's triangle lists:

$$(a + b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k \tag{17}$$

The binomial coefficient counts the subsets with k elements of a set with n elements:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \tag{18}$$

Two choices of a and b in (17) give identities that are worth remembering. With $a = b = 1$ the coefficients add up to 2^n, the number of all subsets:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n \tag{19}$$

With $a = 1$ and $b = -1$ the signs alternate, and for $n \geq 1$ the sum is zero:

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0 \tag{20}$$

8. The number e

The binomial theorem also leads to the number e. Take $b = 1/n$ and $a = 1$ in (17): the coefficient of each power grows towards $1/k!$ as n grows, and in the limit

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{\infty} \frac{1}{k!} \tag{21}$$

so that $e \approx 2.71828$. Both expressions converge, but the series is far faster: its seventh partial sum already has four correct decimals.

9. Checking the numbers

The table lists the partial sums of the series of (1) and their distance to the limit $\pi^2/6 \approx 1.6449$. The error shrinks like $1/n$, and every row respects the bound of (4).

Table 1. Partial sums of the series for ζ(2)

n	$s_n = \sum_{k=1}^{n} 1/k^2$	$\pi^2/6 - s_n$	$2 - 1/n$
1	1.0000	0.6449	1.0000
2	1.2500	0.3949	1.5000
3	1.3611	0.2838	1.6667
4	1.4236	0.2213	1.7500
5	1.4636	0.1813	1.8000
10	1.5498	0.0952	1.9000

Euler’s result, the integral of Gauss and the binomial theorem are three views of the same landscape: sums that look infinite until one finds the right way to fold them.